CHAPTER 2 — ANALYTIC FUNCTIONS

1. Sets of Complex Numbers

1.1. Neighborhood and Points Classification.

ε-Neighborhood. For $z_0 \in \mathbb{C}$, the *ε-neighborhood* of z_0 is the set

$$D_\varepsilon(z_0) = \{z \in \mathbb{C} : |z - z_0| < \varepsilon\}.$$

It is the open disc of radius ε and centered at z_0.

Interior Point, Boundary Point, Isolated Point, Accumulation Point.

A point $z \in E$ is called an *interior point* of set $E \subset \mathbb{C}$ if there exists an ε-neighborhood $D_\varepsilon(z)$ contained in E.

A point $z \in \mathbb{C}$ is called an *exterior point* of set $E \subset \mathbb{C}$ if there exists an ε-neighborhood $D_\varepsilon(z)$ that is disjoint with E.

A point $z \in \mathbb{C}$ is called a *boundary point* of set $E \subset \mathbb{C}$ if it is neither an interior point nor an exterior point of E, that is, any ε-neighborhood $D_\varepsilon(z)$ contains points in E and in complement E^c.

A point $z \in E$ is called an *isolated point* of set $E \subset \mathbb{C}$ if there exist an ε-neighborhood $D_\varepsilon(z)$ that contains no points of E other than z.

A point $z \in \mathbb{C}$ is called an *accumulation point* (limit point) of set $E \subset \mathbb{C}$ if any ε-neighborhood $D_\varepsilon(z)$ contains points of E other than z.

1.2. Open Set and Closed Set.

A subset $E \subset \mathbb{C}$ is called *open* if every point in E is an interior point.

A subset of $\mathbb{C}$ is called *closed* if its complement is open.

The set operations of open and closed sets have the properties:

- Any union of open sets is open, and finite intersection of open sets is open.
- Any intersection of closed set is closed, and finite union of closed set is closed.

Domain. A set $D \subset \mathbb{C}$ is *connected* if any two points $z_1, z_2 \in D$ can be joined by finite number of line segments (a polygonal line). A *domain* is a connected open set.

Example. (**1**) Open disc, $D = \{z \in \mathbb{C} : |z - z_0| < r\}$, $r > 0$.

(**2**) Punctured open disc, $D = \{z \in \mathbb{C} : 0 < |z - z_0| < r\}$.

(**3**) Punctured plane, $D = \{z \in \mathbb{C} : z \neq 0\}$.

(**4**) Annulus, $D = \{z \in \mathbb{C} : 0 < r_0 < |z - z_0| < r\}$.

(**5**) Half-plane, $D = \{z \in \mathbb{C} : \operatorname{Re} z > 0\}$.

(**6**) Branch-cut plane, $D = \mathbb{C} \setminus (-\infty, 0]$.

2. Functions of a Complex Variable

2.1. Complex Functions.

Let $E \subset \mathbb{C}$. A complex-valued *function* $w = f(z)$, $z \in E$, is a relation that for each $z \in E$ there is a unique $w \in \mathbb{C}$ associated with z. The set E is called the *domain of definition* of f.

Example. (**1**) Linear function.

$$f(z) = az + b, \quad z \in \mathbb{C}.$$

(**2**) Reciprocal function. $$f(z) = \frac{1}{z}, \quad z \neq 0.$$

(**3**) Square function.

$$f(z) = z^2, \quad z \in \mathbb{C}.$$

(**4**) Polynomial function.

$$P(z) = a_0 + a_1 z + a_2 z^2 + \cdots + a_n z^n, \quad z \in \mathbb{C}.$$

(**5**) Rational function. $f(z) = \dfrac{P(z)}{Q(z)}$, $Q(z) \neq 0$, where $P(z), Q(z)$ are polynomials.

The graph of a complex function $w = f(z)$ is a surface in a four-dimensional space. We usually consider $f(z)$ as a map from the z-plane to the w-plane.

Let $z = x + iy = re^{i\theta}$ and $w = u + iv$. Then

$$w = f(z) = u(x, y) + iv(x, y), \quad \text{or } w = f(z) = u(r, \theta) + iv(r, \theta).$$

2.2. Transformations.

(**1**) Translation. $w = z + z_0$. $u = x + x_0$, $v = y + y_0$.

(**2**) Rotation. $w = e^{i\theta_0} z$. $\arg w = \arg z + \theta_0$.

(**3**) Dilation. $w = r_0 z$, $r_0 > 0$.

(**4**) Reflection about the x-axis. $w = \bar{z}$.

(**5**) Reflection about the line $\theta = \theta_0$. $w = e^{2i\theta_0} \bar{z}$.

2.3. Examples.

(**1**) Let $w = z^2 = r^2 e^{2i\theta}$. Then $|w| = |z|^2$, $\arg w = 2 \arg z$. Thus, the circle $|z| = r_0$ in the z-plane is the circle $|w| = r_0^2$ in the w-plane. The line $\arg z = \theta_0$ in the x-plane is the line $\arg w = 2\theta_0$ in the w-plane.

We also have $u = x^2 - y^2$, $v = 2xy$. Thus, hyperbola, $x^2 - y^2 = c_1 > 0$, $2xy = c_2 > 0$, in the z-plane, are mapped to straight lines $u = c_1$, $v = c_2$, in the w-plane.

(**2**) Let $w = \dfrac{1}{z}$. Then $|w| = |z|^{-1}$, $\arg w = -\arg z$. Thus, the circle $|z| = r_0$ in the z-plane is the circle $|w| = r_0^{-1}$ in the w-plane. The line $\arg z = \theta_0$ in the x-plane is the line $\arg w = -\theta_0$ in the w-plane.

In terms of x, y coordinates,

$$u = \frac{x}{x^2+y^2}, \quad v = -\frac{y}{x^2+y^2}.$$

For lines $u = c_1 > 0$ and $v = c_2 > 0$ in the w-plane, $\dfrac{x}{x^2+y^2} = c_1$ and $-\frac{y}{x^2+y^2} = c_2$ are broken circles at the origin with radii $\frac{1}{2c_1}$ and $\frac{1}{2c_2}$ and centers $\left(\frac{1}{2c_1}, 0\right)$ and $\left(0, -\frac{1}{2c_2}\right)$.

(**3**) Let $w = z^{\frac{1}{2}}$. For $z = re^{i\theta} \neq 0$, the $z^{\frac{1}{2}}$ has two square roots,

$$f_1(z) = r^{\frac{1}{2}} e^{i\frac{\theta}{2}}, \quad f_2(z) = r^{\frac{1}{2}} e^{i(\frac{\theta}{2}+\pi)} = -r^{\frac{1}{2}} e^{i\frac{\theta}{2}}, \ -\pi < \theta < \pi.$$

The cut z-plane, $\mathbb{C} \setminus (-\infty, 0]$, is mapped to the right-half w-plane, $\{w \in \mathbb{C} : \operatorname{Re} w > 0\}$, by $f_1(z)$, and to the left-half w-plane, $\{w \in \mathbb{C} : \operatorname{Re} w < 0\}$, by $f_2(z)$. The function $w = f_1(z)$ is called the *principal branch* of $w = z^{\frac{1}{2}}$.

2.4. **Limits.**

Definition. Let $f(z)$ be a complex function over E and z_0 is an accumulation point of E. We say that the *limit* of $f(z)$, as z approaches to z_0, is the number w_0, if for any $\varepsilon > 0$, there exists a $\delta > 0$ so that

$$|f(z) - w_0| < \varepsilon, \quad \text{whenever } 0 < |z - z_0| < \delta, \ z \in E,$$

denoted by $\lim\limits_{z \to z_0} f(z) = w_0$. The limit means that $|z - z_0| \to 0 \ \Rightarrow \ |f(z) - w_0| \to 0$.

By using the triangle inequality and the definition of limit, it can be easily shown that the limit of $f(z)$ at z_0 is unique if the limit exists. Indeed, if w_1 is also a limit, then

$$|w_1 - w_0| \leq |f(z) - w_1| + |f(z) - w_0| \to 0 \ \text{ as } z \to z_0,$$

which implies that $w_1 = w_0$.

Let $z - z_0 = re^{i\theta}$. Then $z \to z_0$ means that $|z - z_0| \to 0$, that is, $r \to 0$ uniformly for all θ. Consequently,

$$\lim_{z \to z_0} f(z) = w_0 \quad \text{implies that} \quad \lim_{r \to 0} f(z) = w_0 \quad \text{uniformly for all } \theta.$$

in particular, it implies the directional limit

$$\lim_{r \to 0} f(z_0 + re^{i\theta}) = w_0, \quad \text{for any fixed } \theta.$$

However, existence of directional limit is not sufficient for the existence of $\lim\limits_{z \to z_0} f(z)$.

Example. (**1**) Let $f(z) = a\bar{z}$, $a \in \mathbb{C}$. Show that $\lim\limits_{z \to 1} f(z) = a$.

The case of $a = 0$ is trivial. Assume that $a \neq 0$. For any $\varepsilon > 0$, let $\delta = \varepsilon/|a|$. Then

$$|f(z) - a| = |a\bar{z} - a| = |a||\bar{z} - 1| = |a||z - 1| < \varepsilon, \quad \text{whenever } 0 < |z - 1| < \delta.$$

(**2**) Let $f(z) = \dfrac{z}{|z|}$. Show that $\lim\limits_{z \to 0} f(z)$ does not exist.

Let $z = re^{i\theta}$. $z \to 0$ means that $|z| = r \to 0$. Then $\dfrac{z}{|z|} = e^{i\theta}$, which does not have a limit when $r \to 0$ because of the uniqueness of limit.

2.5. **Properties of Limits.**

Proposition 2.1. *Let $z = x + iy$, $w = f(z) = u + iv$, $z_0 = x_0 + iy_0$, and $w_0 = u_0 + iv_0$. Then*

$$\lim_{z\to z_0} f(z) = w_0 \quad \textit{if and only if} \quad \lim_{(x,y)\to(x_0,y_0)} u(x,y) = u_0, \quad \lim_{(x,y)\to(x_0,y_0)} v(x,y) = v_0.$$

Proof. "If part". By the inequalities,

$$\max\{|u-u_0|, |v-v_0|\} \le |w-w_0| \le |u-u_0| + |v-v_0|, \ \max\{|x-x_0|, |y-y_0|\} \le |z-z_0| \le |x-x_0| + |y-y_0|,$$

we have

$$|x-x_0| \to 0, \ |y-y_0| \to 0 \ \Rightarrow \ |z-z_0| \to 0 \ \Rightarrow \ |w-w_0| \to 0 \ \Rightarrow \ |u-u_0| \to 0, \ |v-v_0| \to 0.$$

"Only if part". Again by the inequalities,

$$|z-z_0| \to 0 \ \Rightarrow \ |x-x_0| \to 0, \ |y-y_0| \to 0 \ \Rightarrow \ |u-u_0| \to 0, \ |v-v_0| \to 0 \ \Rightarrow \ |w-w_0| \to 0.$$

□

Proposition 2.2. *If*

$$\lim_{z\to z_0} f(z) = w_0 \quad \textit{and} \quad \lim_{z\to z_0} g(z) = w_1,$$

then

$$\lim_{z\to z_0} (f(z) + g(z)) = w_0 + w_1, \quad \lim_{z\to z_0} f(z)g(z) = w_0 w_1,$$

and

$$\lim_{z\to z_0} \frac{f(z)}{g(z)} = \frac{w_0}{w_1}, \quad \textit{when } w_1 \neq 0.$$

This proposition can be shown easily by using Proposition 2.1.

Example. (**1**) Let $P(z)$ and $Q(z)$ be polynomials. If $Q(z_0) \neq 0$, then

$$\lim_{z\to z_0} \frac{P(z)}{Q(z)} = \frac{P(z_0)}{Q(z_0)}.$$

If $Q(z_0) = 0$ and $P(z_0) \neq 0$, then $\lim_{z\to z_0} \frac{P(z)}{Q(z)}$ does not exist.

(**2**) Calculate the limits

$$\lim_{z\to i} \frac{z+i}{z^4 + iz^3 - z - i} = -\frac{1}{i+1}, \quad \lim_{z\to 1} \frac{z+i}{z^4 + iz^3 - z - i} \text{ does not exist.}$$

$$\lim_{z\to -i} \frac{z+i}{z^4 + iz^3 - z - i} = \lim_{z\to -i} \frac{z+i}{(z+i)(z^3-1)} = \frac{1}{i-1}.$$

$$\lim_{z\to -i} \frac{(z+i)^2}{z^4 + iz^3 - z - i} = 0, \quad \lim_{z\to -i} \frac{z+i}{(z^4 + iz^3 - z - i)^2} \text{ does not exist.}$$

2.6. **Limit at Infinity.**

Suppose the domain E of $f(z)$ is unbounded. We say that $f(z)$ has limit w_0 as $z \to \infty$ if for any $\varepsilon > 0$, there exists $N > 0$ so that

$$|f(z) - w_0| < \varepsilon, \quad \text{whenever } |z| > N, z \in E,$$

denoted by $\lim_{z\to\infty} f(z) = w_0$.

Proposition 2.3.

$$\lim_{z\to\infty} f(z) = w_0 \quad \textit{if and only if} \quad \lim_{z\to 0} f\Big(\frac{1}{z}\Big) = w_0.$$

Example. Let $P(z) = a_0 + a_1 z + \cdots + a_n z^n$ and $Q(z) = b_0 + b_1 z + \cdots + b_n z^n$, $b_n \neq 0$. Then

$$\lim_{z\to\infty} \frac{P(z)}{Q(z)} = \lim_{z\to 0} \frac{P(1/z)}{Q(1/z)} = \lim_{z\to 0} \frac{a_0 z^n + a_1 z^{n-1} + \cdots + a_n}{b_0 z^n + b_1 z^{n-1} + \cdots + b_n} = \frac{a_n}{b_n}.$$

2.7. **Continuity.**

Definition. A complex function $f(z)$ over E is said to be *continuous* at $z_0 \in E$ if

$$\lim_{z\to z_0} f(z) = f(z_0),$$

that is,

$$z \to z_0 \;\Rightarrow\; f(z) \to f(z_0).$$

If $f(z)$ is continuous at every point of E, then $f(z)$ is called a continuous function on E.

By Proposition 2.2, we have

Proposition 2.4. *If $f(z)$ and $g(z)$ are continuous functions on E, then*

(**1**) *$f(z) + g(z)$ and $f(z)g(z)$ are continuous on E;*

(**2**) *$\dfrac{f(z)}{g(z)}$ is continuous on E when $g(z) \neq 0$.*

Since linear functions are clearly continuous on $\mathbb{C}$, this proposition implies that any polynomial $P(z)$ is continuous on $\mathbb{C}$, and any rational function $\dfrac{P(z)}{Q(z)}$ is continuous on its domain where $Q(z) \neq 0$.

Proposition 2.5. *A composition of continuous functions is continuous.*

Proof. Suppose $w = f(z)$ is continuous at z_0, $w_0 = f(z_0)$, and $s = g(w)$ is continuous at w_0. We need to show that $g(f(z))$ is continuous at z_0. By the continuity of f and g, we have

$$z \to z_0 \;\Rightarrow\; w = f(z) \to w_0 \;\Rightarrow\; g(w) \to g(w_0) = g(f(z_0)).$$

This can be easily rewritten in terms of $\varepsilon - \delta$. □

Proposition 2.6. *If a complex function $f(z)$ is continuous at z_0 and $f(z_0) \neq 0$, then the exists a δ-neighborhood $D_\delta(z_0)$ so that $f(z) \neq 0$ when $z \in D_\varepsilon(z_0)$.*

Proof. Since $f(z)$ is continuous at z_0, for any $\varepsilon > 0$, there exists $\delta > 0$ so that

$$|f(z) - f(z_0)| < \varepsilon, \quad \text{whenever } z \in D_\delta(z_0).$$

Let $\varepsilon = \frac{1}{2}|f(z_0)|$. Then

$$|f(z_0)| - |f(z)| \le |f(z) - f(z_0)| < \frac{1}{2}|f(z_0)|, \quad z \in D_\delta(z_0),$$

and thus,

$$0 < \frac{1}{2}|f(z_0)| \le |f(z)|, \quad z \in D_\delta(z_0).$$

□

Theorem 2.7. *Let $E \subset \mathbb{C}$ be closed and bounded and let $f(z)$ be a continuous function on E. Then there exists $z_0 \in E$ so that*

$$|f(z)| \le |f(z_0)|, \quad z \in E.$$

Proof. Let $B = \{|f(z)| : z \in E\}$. Then $B \subset [0, \infty)$. We show that B is closed and bounded in $\mathbb{R}$. Suppose B is not bounded. Then there is a sequence $z_n \in E$ so that $|f(z_n)| \to \infty$. By the Bolzano-Weierstrass theorem, the bounded sequence $\{z_n\}$ has a convergent subsequence $z_{n_k} \to z' \in E$. Since $f(z)$ is continuous at z', $|f(z_{n_k})| \to |f(z')| < \infty$, a contradiction.

To show that B is closed, let $p_n \in B$ and $p_n \to p$. We need to show $p \in B$.

There are z_n so that $p_n = |f(z_n)|$. Bounded sequence $\{z_n\}$ has a convergent subsequence $\{z_{n_k}\}$. Let $z_{n_k} \to z'$. Then $z' \in E$ because E is closed. Thus, $p_{n_k} = |f(z_{n_k})| \to f(z')$ because $f(z)$ is continuous at z'. Therefore, $p = |f(z')| \in B$.

Since B is bounded and closed, $\sup B$ exists and $\sup B \in B$. Thus, there exists $z_0 \in E$ so that $\sup B = |f(z_0)|$. □

3. Complex Derivative

3.1. Derivative.

Definition. Let $w = f(z)$ be a complex function on a domain D. For $z_0 \in D$, the *derivative* of f at z_0 is the limit,

$$f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0}.$$

The function $f(z)$ is called *differentiable* at z_0 if $f'(z_0)$ exists. The function $f(z)$ is called *differentiable* on D if it is differentiable at every point $z \in D$. Then $f'(z)$ is a complex function on D. The function $f(z)$ is called *continuously differentiable* on D if the derivative function $f'(z)$ is continuous on D.

It is also written,

$$f'(z_0) = \lim_{\lambda \to 0} \frac{f(z_0 + \lambda) - f(z_0)}{\lambda} \quad \text{or} \quad f'(z_0) = \lim_{\Delta z \to 0} \frac{\Delta w}{\Delta z},$$

where $\Delta z = z - z_0$ and $\Delta w = f(z_0 + \Delta z) - f(z_0)$.

Example. (**1**) The function $w = f(z) = z^n$ is differentiable for any z, where n is a positive integer.

$$f'(z) = \lim_{\lambda \to 0} \frac{(z + \lambda)^n - z^n}{\lambda} = \lim_{\lambda \to 0} \frac{nz^{n-1}\lambda + \cdots + \lambda^n}{\lambda} = nz^{n-1}.$$

(**2**) The function $w = f(z) = \bar{z}$ is not differentiable anywhere.

$$\lim_{\lambda \to 0} \frac{\overline{z + \lambda} - \bar{z}}{\lambda} = \lim_{\lambda \to 0} \frac{\bar{\lambda}}{\lambda},$$

which has no limit as shown in Section 2.4 (2).

(**3**) The function $w = f(z) = z\bar{z}$ is differentiable at $z = 0$ only.

$$\lim_{\lambda\to 0}\frac{(z+\lambda)(\overline{z+\lambda})-z\bar{z}}{\lambda} = \lim_{\lambda\to 0}\frac{\lambda\bar{z}+\bar{\lambda}z+\lambda\bar{\lambda}}{\lambda} = \lim_{\lambda\to 0}\Big(\bar{z}+\frac{\bar{\lambda}}{\lambda}z+\bar{\lambda}\Big),$$

which has a limit at $z = 0$ only.

Differentiability implies continuity.

$$\lim_{z\to z_0}(f(z)-f(z_0)) = \lim_{z\to z_0}\frac{f(z)-f(z_0)}{z-z_0}\lim_{z\to z_0}(z-z_0) = f'(z_0)\cdot 0 = 0.$$

3.2. **Derivative Formulas.**

Proposition 3.1. *Let $f(z)$ and $g(z)$ be complex functions which are differentiable at z, then*

(**1**) *$f+g$ is differentiable at z, and $(f+g)'(z) = f'(z)+g'(z)$;*
(**2**) *fg is differentiable at z, and $(fg)'(z) = f'(z)g(z)+f(z)g'(z)$;*
(**3**) *$\dfrac{f}{g}$ is differentiable at z when $g(z)\neq 0$, and $\Big(\dfrac{f}{g}\Big)'(z) = \dfrac{f'(z)g(z)-f(z)g'(z)}{g^2(z)}$;*

By this proposition, any polynomial $P(z)$ is differentiable on $\mathbb{C}$, and any rational function $\dfrac{P(z)}{Q(z)}$ is differentiable on its domain where $Q(z)\neq 0$.

Proposition 3.2. *If $w = f(z)$ is differentiable at z and $g(w)$ is differentiable at w, then the composition $(g\circ f)(z) = g(f(z))$ is differentiable at z and $(g\circ f)'(z) = g'(f(z))f'(z)$.*

Proof. Let $w_0 = f(z_0)$ and w be close to w_0. Consider the function $G(w)$,

$$G(w) = \begin{cases}\frac{g(w)-g(w_0)}{w-w_0} & w\neq w_0\\ g'(w_0) & w = w_0\end{cases}.$$

Then $G(w)$ is continuous at w_0. When $f(z)\neq f(z_0)$, we have

$$\frac{g(f(z))-g(f(z_0))}{z-z_0} = G(f(z))\frac{f(z)-f(z_0)}{z-z_0}$$

This identity still holds when $f(z) = f(z_0)$ because it becomes $0 = g'(w_0)\cdot 0$.

Since $w = f(z)$ is continuous at z_0 and $G(w)$ is continuous at w_0, we have $G(f(z))\to g'(f(z_0))$ as $z\to z_0$. Thus, taking limit $z\to z_0$ gives

$$(g\circ f)'(z_0) = g'(f(z_0))f'(z_0).$$

□

Example. Let $f(z) = \Big(\dfrac{1-z^2}{1+z^2}+iz^2\Big)^3$. Find $f'(z)$.

$$f'(z) = 3\Big(\frac{1-z^2}{1+z^2}+iz^2\Big)^2\Big(\frac{-2z(1+z^2)-2z(1-z^2)}{(1+z^2)^2}+2iz\Big) = 6z\Big(\frac{1-z^2}{1+z^2}+iz^2\Big)^2\Big(\frac{-2}{(1+z^2)^2}+i\Big).$$

4. Cauchy-Riemann Equations

4.1. Cauchy-Riemann Equations in Cartesian Coordinates.

Let $w = f(z) = u + iv$ be a complex function differentiable in a domain D. We calculate the derivative

$$f'(z) = \lim_{\Delta z \to 0} \frac{f(z + \Delta z) - f(z)}{\Delta z}.$$

Take the limit $\Delta z \to 0$ along the x-axis and y-axis, that is, $\Delta z = \Delta x \to 0$ and $\Delta z = i\Delta y \to 0$.

For $\Delta z = \Delta x \to 0$, we have

$$\begin{aligned}\frac{f(z + \Delta x) - f(z)}{\Delta x} &= \frac{u(x + \Delta x, y) + iv(x + \Delta x, y) - (u(x, y) + iv(x, y))}{\Delta x} \\ &= \frac{u(x + \Delta x, y) - u(x, y)}{\Delta x} + i\frac{v(x + \Delta x, y) - v(x, y)}{\Delta x},\end{aligned}$$

and

$$f'(z) = \frac{\partial u}{\partial x}(x, y) + i\frac{\partial v}{\partial x}(x, y) = \frac{\partial f}{\partial x}, \quad z = x + iy.$$

For $\Delta z = i\Delta y \to 0$, we have

$$\begin{aligned}\frac{f(z + i\Delta y) - f(z)}{i\Delta y} &= \frac{u(x, y + \Delta y) + iv(x, y + \Delta y) - (u(x, y) + iv(x, y))}{\Delta x} \\ &= \frac{v(x, y + \Delta y) - v(x, y)}{\Delta y} - i\frac{u(x, y + \Delta y) - u(x, y)}{\Delta y},\end{aligned}$$

and

$$f'(z) = \frac{\partial v}{\partial y}(x, y) - i\frac{\partial u}{\partial y}(x, y) = -i\frac{\partial f}{\partial y}, \quad z = x + iy.$$

This calculation gives

Cauchy-Riemann equations.

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}, \quad \text{or} \quad \frac{\partial f}{\partial x} = -i\frac{\partial f}{\partial y}. \tag{4.1}$$

One can also write the equations as the following,

$$J = \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{pmatrix} = \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ -\frac{\partial u}{\partial y} & \frac{\partial u}{\partial x} \end{pmatrix} = \begin{pmatrix} \frac{\partial v}{\partial y} & -\frac{\partial v}{\partial x} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{pmatrix}.$$

The matrix J satisfies the identity $J + J^t = \lambda I$, where I is the identity matrix and $\lambda = 2u_x = 2v_y$. Also, note that the row vectors are orthogonal.

Example. The function $f(z) = |z|^2$ is not differentiable when $z \neq 0$.

$$|z|^2 = u + iv = x^2 + y^2, \quad u = x^2 + y^2, \quad v = 0.$$

$$\frac{\partial u}{\partial x} = 2x, \quad \frac{\partial v}{\partial y} = 0$$

Thus, the Cauchy-Riemann equations are not satisfied when $z \neq 0$, and hence not differentiable.

Theorem 4.1. *Let $w = f(z) = u(x,y) + iv(x,y)$ be a complex function defined on a domain D. Then $f(z)$ is continuously differentiable on D if and only if $u(x,y)$ and $v(x,y)$ have continuous first order partial derivatives satisfying the Cauchy-Riemann equations.*

Proof. The necessary part has been proved above. We show the sufficient part. Since u and v have continuous first order partial derivatives, we have

$$\Delta u = \frac{\partial u}{\partial x}(x,y)\Delta x + \frac{\partial u}{\partial y}(x,y)\Delta y + \varepsilon_3\Delta x + \varepsilon_4\Delta y$$
$$\Delta v = \frac{\partial v}{\partial x}(x,y)\Delta x + \frac{\partial v}{\partial y}(x,y)\Delta y + \varepsilon_3\Delta x + \varepsilon_4\Delta y,$$

where ε_k, $k = 1,\dots,4$, tend to 0 as Δx and Δy tend to 0. Then by the Cauchy-Riemann equations,

$$\begin{aligned}\Delta w =& \frac{\partial u}{\partial x}\Delta x + \frac{\partial u}{\partial y}\Delta y + \varepsilon_1\Delta x + \varepsilon_2\Delta y + i\Big(\frac{\partial v}{\partial x}\Delta x + \frac{\partial v}{\partial y}\Delta y + \varepsilon_3\Delta x + \varepsilon_4\Delta y\Big)\\ =& \frac{\partial u}{\partial x}\Delta x - \frac{\partial v}{\partial x}\Delta y + \varepsilon_1\Delta x + \varepsilon_2\Delta y + i\Big(\frac{\partial v}{\partial x}\Delta x + \frac{\partial u}{\partial x}\Delta y + \varepsilon_3\Delta x + \varepsilon_4\Delta y\Big)\\ =& \Big(\frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x}\Big)(\Delta x + i\Delta y) + (\varepsilon_1 + i\varepsilon_3)\Delta x + (\varepsilon_2 + i\varepsilon_4)\Delta y.\end{aligned}$$

Thus,

$$\frac{\Delta w}{\Delta z} = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} + (\varepsilon_1 + i\varepsilon_3)\frac{\Delta x}{\Delta z} + (\varepsilon_2 + i\varepsilon_4)\frac{\Delta y}{\Delta z}.$$

Since

$$\Big|(\varepsilon_1 + i\varepsilon_3)\frac{\Delta x}{\Delta z}\Big| \le |\varepsilon_1| + |\varepsilon_3|, \quad \Big|(\varepsilon_2 + i\varepsilon_4)\frac{\Delta x}{\Delta z}\Big| \le |\varepsilon_2| + |\varepsilon_4|,$$

We obtain

$$f'(z) = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x},$$

The derivative function $f'(z)$ is continuous because the partial derivatives are continuous. □

Theorem 4.1 is formulated in a domain. One can also state the result at a fixed point.

Theorem 4.2. *Let $w = f(z) = u(x,y) + iv(x,y)$ be a complex function defined in a neighborhood of point $z_0 = x_0 + iy_0$.*

(**1**) *If $f'(z_0)$ exists, then the first order partial derivatives of u and v exist at (x_0, y_0) and satisfy the Cauchy-Riemann equations.*

(**2**) *If the first order partial derivatives of u and v exist in a neighborhood of (x_0, y_0), are continuous at (x_0, y_0), and satisfy the Cauchy-Riemann equations at (x_0, y_0), then $f'(z_0)$ exists.*

The proof of Theorem 4.2 is the same as that of Theorem 4.1.

Example. (**1**) Let $f(z)$ be the exponential function,

$$f(z) = e^z = e^x e^{iy} = e^x(\cos y + i\sin y), \quad z = x + iy.$$

Then $f'(z) = f(z)$.

We have

$$f'(z) = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} = e^x \cos y + ie^x \sin y = e^z = f(z).$$

(**2**) Let $f(z) = e^{P(z)}$, where $P(z)$ is a polynomial. Then

$$f'(z) = f(z)P'(z).$$

(**3**) Let $f(z) = \frac{x^4}{4} + i\frac{y^2}{2}$. determine where $f'(z)$ exists and find its value.

We have

$$u_x = x^3, \quad u_y = 0, \quad v_x = 0, \quad v_y = y.$$

By Cauchy-Riemann equations, $u_x = v_y$ gives $y = x^3$ and $u_y = -v_x = 0$. Thus

$$f'(z) \text{ exists on } y = x^3, \text{ and } f'(x + ix^3) = x^3.$$

4.2. **Cauchy-Riemann Equations in Polar Coordinates.**

Let

$$z = x + iy = r\cos\theta + ir\sin\theta = re^{i\theta}, \quad z \neq 0,$$

and

$$w = f(z) = u + iv = u(x, y) + iv(x, y) = u(r, \theta) + iv(r, \theta).$$

We have

$$\begin{aligned}
\frac{\partial u}{\partial r} &= \frac{\partial u}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial r} = \frac{\partial u}{\partial x}\cos\theta + \frac{\partial u}{\partial y}\sin\theta, \\
\frac{\partial u}{\partial \theta} &= \frac{\partial u}{\partial x}\frac{\partial x}{\partial \theta} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial \theta} = -\frac{\partial u}{\partial x}r\sin\theta + \frac{\partial u}{\partial y}r\cos\theta.
\end{aligned} \tag{4.2}$$

Similarly,

$$\begin{aligned}
\frac{\partial v}{\partial r} &= \frac{\partial v}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial v}{\partial y}\frac{\partial y}{\partial r} = \frac{\partial v}{\partial x}\cos\theta + \frac{\partial v}{\partial y}\sin\theta, \\
\frac{\partial v}{\partial \theta} &= \frac{\partial v}{\partial x}\frac{\partial x}{\partial \theta} + \frac{\partial v}{\partial y}\frac{\partial y}{\partial \theta} = -\frac{\partial v}{\partial x}r\sin\theta + \frac{\partial v}{\partial y}r\cos\theta.
\end{aligned} \tag{4.3}$$

By the Cauchy-Riemann equations (4.1), equations (4.3) become

$$\frac{\partial v}{\partial r} = -\frac{\partial u}{\partial y}\cos\theta + \frac{\partial u}{\partial x}\sin\theta, \quad \frac{\partial v}{\partial \theta} = \frac{\partial u}{\partial y}r\sin\theta + \frac{\partial u}{\partial x}r\cos\theta. \tag{4.4}$$

By (4.2) and (4.4), we have

Cauchy-Riemann Equations.

$$r\frac{\partial u}{\partial r} = \frac{\partial v}{\partial \theta}, \quad \frac{\partial u}{\partial \theta} = -r\frac{\partial v}{\partial r}, \quad \text{or} \quad \frac{\partial f}{\partial r} = \frac{1}{ir}\frac{\partial f}{\partial \theta}. \tag{4.5}$$

We rewrite (4.2),

$$\begin{pmatrix} \frac{\partial u}{\partial r} \\ \frac{\partial u}{\partial \theta} \end{pmatrix} = \begin{pmatrix} \cos\theta & \sin\theta \\ -r\sin\theta & r\cos\theta \end{pmatrix} \begin{pmatrix} \frac{\partial u}{\partial x} \\ \frac{\partial u}{\partial y} \end{pmatrix}.$$

Finding the inverse matrix and using the Cauchy-Riemann equations, we have

$$\begin{pmatrix} \frac{\partial u}{\partial x} \\ -\frac{\partial v}{\partial x} \end{pmatrix} = \frac{1}{r}\begin{pmatrix} r\cos\theta & -\sin\theta \\ r\sin\theta & \cos\theta \end{pmatrix}\begin{pmatrix} \frac{\partial u}{\partial r} \\ -r\frac{\partial v}{\partial r} \end{pmatrix}$$

Thus,

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r}\cos\theta + \frac{\partial v}{\partial r}\sin\theta, \quad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial r}\sin\theta + \frac{\partial v}{\partial r}\cos\theta,$$

which gives the

Derivative Formula.

$$f'(z) = \frac{\partial f}{\partial x} = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} = (\cos\theta - i\sin\theta)\Big(\frac{\partial u}{\partial r} + i\frac{\partial v}{\partial r}\Big) = e^{-i\theta}\frac{\partial f}{\partial r}. \tag{4.6}$$

Example. Let $f(z) = z^{\frac{1}{n}}$ and

$$f(z) = r^{\frac{1}{n}}e^{i\frac{\theta}{n}}, \quad r > 0,\ \alpha < \theta < \alpha + 2\pi,\ \alpha \in \mathbb{R}.$$

Then

$$f'(z) = e^{-i\theta}\frac{1}{n}r^{\frac{1-n}{n}}e^{i\frac{\theta}{n}} = \frac{1}{n}r^{\frac{1-n}{n}}e^{i\frac{(1-n)\theta}{n}} = \frac{1}{n(r^{\frac{1}{n}}e^{i\frac{\theta}{n}})^{n-1}} = \frac{1}{n(f(z))^{n-1}}.$$

5. Analytic Functions

5.1. Definition of Analytic.

A complex function $f(z)$ is called *analytic at a point* z_0 if $f(z)$ is differentiable in a neighborhood of z_0.

A complex function $f(z)$ is called *analytic in an open set* U if $f(z)$ is differentiable at every point of U.

A complex function $f(z)$ is called *analytic in a set* E if $f(z)$ is analytic in some open set $U \supset E$.

Example.

(**1**) The function $f(z) = z^2$ is analytic in $\mathbb{C}$.

(**2**) The function $f(z) = \dfrac{1}{z}$ is analytic in the domain $\mathbb{C} \setminus \{0\}$.

(**3**) The function $f(z) = \dfrac{z+i}{(z-i)(z^2-1)}$ is analytic in the domain $\mathbb{C} \setminus \{i, 1, -1\}$.

(**4**) The function $f(z) = |z|^2$ is not analytic anywhere because it is differentiable at $z = 0$ only.

(**5**) The function $f(z) = \bar{z}$ is not analytic anywhere because it is not differentiable anywhere.

Proposition 5.1. *Let f and g be analytic functions. Then $f + g$ and fg are analytic, and f/g is analytic when $g \neq 0$.*

5.2. Entire Function and Singular Point.

An *entire function* is a complex function that is differentiable in $\mathbb{C}$.

A polynomial $P(z)$ is an entire function. The exponential function e^z is an entire function.

A point z_0 is called a *singular point* of $f(z)$ if $f(z)$ is not analytic at z_0 but is analytic at some point in every ε-neighborhood of z_0.

Example.

(**1**) The point $z = 0$ is a singular point of $f(z) = \frac{1}{z}$.

(**2**) The function $f(z) = |z|^2$ has no singular point because it has no analytic point.

Theorem 5.2. *If $f'(z) = 0$ in a domain D, then $f(z)$ is constant in D.*

Proof. Let $f(z) = u(x, y) + iv(x, y)$, $z = x + iy$. We have

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial y} = 0, \quad \frac{\partial v}{\partial x} = \frac{\partial v}{\partial y} = 0.$$

For two points P, P' in D, suppose the line segment L joining P and P' is contained in D. We show that u is constant on L. Parameterize a point (x, y) on L by its distance s to P. Then, we can write $u = u(x(s), y(s))$. We have

$$\frac{du}{ds} = \frac{\partial u}{\partial x}\frac{dx}{ds} + \frac{\partial u}{\partial y}\frac{dy}{ds} = 0.$$

This means that u constant on L.

Since D is a domain, any two points P and Q in D can be joined by finitely many line segments in D. Then u is constant on the polygonal line and thus has the same value at P and Q, that is, u is constant in D. Similarly, v is also constant in D. □

Example.

(**1**) Let

$$f(z) = \frac{az + b}{cz + d}, \quad ad - bc \neq 0, \ a, b, c, d \in \mathbb{R}.$$

The singular point of f is $z = -d/c$. The function f is analytic in the domain $D = \mathbb{C} \setminus \{-d/c\}$. Its inverse function is

$$z = f^{-1}(w) = \frac{dw - b}{-cw + a},$$

which is analytic in the domain $\mathbb{C} \setminus \{a/c\}$.

(**2**) Let

$$f(z) = \frac{z^3 + 3}{(z^2 - 2)(z^2 + 1)}.$$

The singular points of $f(z)$ are $z = \pm\sqrt{2}$ and $z = \pm i$. The function f is analytic in the domain $D = \mathbb{C} \setminus \{\pm\sqrt{2}, \pm i\}$.

Lemma 5.3. *Suppose both $f(z)$ and $\overline{f(z)}$ are analytic in a domain D. Show that $f(z)$ is constant in D.*

Proof. Let $f = u + iv$ and $\bar{f} = u - iv$. By the Cauchy-Riemann equations, we have

$$u_x = v_y, \ u_y = -v_x; \quad u_x = -v_y, \ u_y = v_x.$$

These equations give

$$u_x = u_y = v_x = v_y = 0,$$

and hence $f'(z) = 0$ in D. By Theorem 5.2, f is constant in D. □

This lemma implies that a real-valued analytic function is constant.

Theorem 5.4. *Let f be an analytic function in a domain D. If the modulus $|f(z)|$ is constant in D, then $f(z)$ is constant in D.*

Proof. Let $|f(z)| = c$, for all $z \in D$, where c is a nonegative constant. When $c = 0$, we have $f(z) = 0$ in D. When $c > 0$, we have $f(z)\overline{f(z)} = c^2 > 0$. Thus, $f(z) \neq 0$, and

$$\overline{f(z)} = \frac{c^2}{f(z)}, \quad z \in D.$$

This implies that $\overline{f(z)}$ is analytic. By Lemma 5.3, $f(z)$ is constant in D. □

6. Harmonic Functions

Let $H(x, y)$ be a real-valued function defined over a domain D in $\mathbb{R}^2$. The function H is said to be *harmonic* in D if it satisfies the *Laplace equation*,

$$H_{xx} + H_{yy} = 0.$$

Theorem 6.1. *Let $f(z) = u(x, y) + iv(x, y)$ be an analytic function in a domain D. Then both u and v are harmonic in D.*

Proof. By the Cauchy-Riemann equations,

$$u_x = v_y, \ u_y = -v_x.$$

Differentiating with respect to x gives,

$$u_{xx} = v_{yx}, \ u_{yx} = -v_{xx}.$$

Differentiating with respect to y gives,

$$u_{xy} = v_{yy}, \ u_{yy} = -v_{xy}.$$

We use the fact that $u_{yx} = u_{xy}$ and $v_{yx} = v_{xy}$. Then

$$u_{xx} + u_{yy} = 0, \quad v_{xx} + v_{yy} = 0.$$

Thus, both u and v are harmonic. □

Example. (**1**) Let $f(z) = e^{-y}\sin x - ie^{-y}\cos x$. Then

$$f_x = e^{-y}\cos x + ie^{-y}\sin x = -if_y.$$

This is the Cauchy-Riemann equation. f is analytic in $\mathbb{C}$, and hence $e^{-y}\sin x$ and $e^{-y}\cos x$ are harmonic in $\mathbb{C}$.

(**2**) Let $f(z) = \frac{i}{z^2}$. f is analytic in $\mathbb{C} \setminus \{0\}$. We have

$$\frac{i}{z^2} = \frac{i\bar{z}^2}{|z|^4} = \frac{2xy + i(x^2 - y^2)}{(x^2 + y^2)^2}.$$

Then

$$u(x, y) = \frac{2xy}{(x^2 + y^2)^2}, \quad v(x, y) = \frac{x^2 - y^2}{(x^2 + y^2)^2}$$

are harmonic in $\mathbb{C} \setminus \{0\}$.

Harmonic Conjugate.

If u is harmonic in a domain D, and v is harmonic in D so that $u + iv$ is analytic, then v is a *harmonic conjugate* of u.

Given a harmonic function, the harmonic conjugate v of u is unique, up to a constant. Indeed, suppose v_1 is another harmonic conjugate of u. Then $u + iv_1$ is also analytic. Thus, the difference $u + iv - (u + iv_1) = i(v - v_1)$ is analytic. Consequently, $v - v_1$ is a real-valued analytic function, and hence is constant.

Example.

(**1**) Given harmonic function $u(x, y) = xy$, find a harmonic conjugate of u.

It is easily seen that u is harmonic. To find a harmonic conjugate v, we use the Cauchy-Riemann equations. By

$$u_x = y = v_y,$$

we have

$$v(x, y) = \frac{y^2}{2} + h(x).$$

By the Cauchy-Riemann equation

$$u_y = x = -v_x,$$

we obtain

$$x = -h'(x), \quad \text{and} \quad h(x) = -\frac{x^2}{2} + C.$$

Thus,

$$v(x, y) = \frac{y^2}{2} - \frac{x^2}{2} + C,$$

and

$$f(z) = u + iv = -i\frac{z^2}{2} + iC.$$

(**2**) Let $u = \cosh x \cos y$. Show that u is harmonic and find its harmonic conjugate v.

Obviously,

$$u_{xx} = \cosh x \cos y = -u_{yy},$$

and u is harmonic.

By the Cauchy-Riemann equations,

$$u_x = \sinh x \cos y = v_y,$$

we have

$$v(x, y) = \sinh x \sin y + h(x).$$

By the Cauchy-Riemann equation

$$u_y = -\cosh x \sin y = -v_x,$$

we obtain

$$h'(x) = 0, \quad \text{and} \quad h(x) = C.$$

Thus,

$$v(x, y) = \sinh x \sin y + C.$$

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